

Polish Power System's Dynamics

A Hybrid EMD-SARIMA Approach for High-Frequency Load Forecasting

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Abstract

Accurate load forecasting is critical for the stability and economic efficiency of national power grids. However, modeling high-frequency electricity demand remains challenging due to the inherent non-stationarity, multi-scale seasonality, and complex socio-economic drivers embedded within the power system. This study aims to develop a robust hybrid forecasting framework capable of decoupling macroscopic deterministic trends from short-term stochastic volatility, thereby improving the accuracy and reliability of multi-step-ahead electricity demand predictions for the Polish national grid.

The proposed methodology relies on a three-stage signal decoupling strategy. First, Empirical Mode Decomposition (EMD) is applied to extract the non-linear, long-term macroeconomic baseline. Second, a seasonal folding technique is utilized to isolate the deterministic weekly profile for each meteorological season, capturing rigid anthropogenic routines. Finally, the isolated, strictly stationary, zero-mean residuals are modeled using an algorithmically optimized SARIMA framework.

The hybrid architecture demonstrates high predictive accuracy over a 15-day out-of-sample horizon during standard operational periods. By restricting the SARIMA algorithm to purely stationary signal, the method successfully prevents the divergence of forecast variance, yielding bounded and stable 95% confidence intervals over time. Nevertheless, the evaluation of anomalous periods reveals significant forecast degradation during unmodeled exogenous events, such as the Christmas holidays and the severe February 2012 cold wave, or when predictions are extended over longer time horizons that span across seasonal transitions.

The EMD-SARIMA framework provides a theoretically rigorous and highly effective baseline for capturing grid dynamics. To manage the non-deterministic volatility of the grid entirely, future iterations must integrate exogenous weather covariates and holiday-aware logic.

Keywords: Time series, Power load, Forecasting, EMD, SARIMA

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1. Introduction

The accurate forecasting of electricity demand is a fundamental pillar for the strategic planning and operational management of national power grids. For utility companies and Transmission System Operators (TSOs), precise load estimates are crucial not only to ensure grid stability and optimal unit commitment but also to minimize operational costs and prevent severe power outages [Misiurek et al., 2025]. Furthermore, as modern power systems transition towards sustainability, robust demand forecasting becomes essential to balancing the intermittent nature of renewable energy sources and to sizing future energy infrastructure.

However, forecasting electricity consumption based on historical data is a highly non-trivial task. Power load time series are inherently complex, non-stationary signals driven by a multitude of anthropogenic and environmental factors. They exhibit strong multi-scale seasonality—closely mirroring human daily and weekly socio-economic routines—alongside non-linear, long-term trends. Additionally, these series are subject to a significant stochastic component driven by exogenous shocks, such as sudden economic downturns, energy crises, or extreme meteorological events (e.g., exceptionally cold winters or heatwaves).

In this study, we analyze the historical electricity load of the Polish power system spanning the period from 2008 to 2016, a grid characterized by distinct thermal-driven seasonality and ongoing structural transitions [Zielińska-Sitkiewicz et al., 2021]. To address the inherent complexity of the signal, we propose a comprehensive analytical framework that systematically decouples the deterministic components from the stochastic noise.

The remainder of this paper is organized as follows: Section 2 introduces the dataset and out-

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lines the preliminary data pre-processing steps. In Section 2.2, we apply Empirical Mode Decomposition (EMD) to isolate and extract the non-linear, long-term trend of the series. Section 3 details the deseasonalization process, where Spectral Analysis via Fast Fourier Transform (FFT) (3.1) and folding techniques (3.2) are employed to identify and extract the dominant human-driven periodicities (i.e., the typical weekly consumption profile). By subtracting both the global trend and the seasonal profile from the original signal, we obtain stationary stochastic residuals. In Section 4, we implement a hybrid forecasting strategy by training a Seasonal Autoregressive Integrated Moving Average (SARIMA) model exclusively on these isolated residuals. Finally, Section 4.3 presents our forecasting results, and highlights the limitations of the proposed architecture by evaluating its performance during anomalous periods, such as national holidays and abnormally severe winter events.

2. Data preparation and Trend extraction

The empirical data utilized in this study consists of the historical electrical load of the Polish power system, obtained from the official open-data repository of the Polish Transmission System Operator, Polskie Sieci Elektroenergetyczne (PSE) [Polskie Sieci Elektroenergetyczne (PSE), 2017]. The dataset comprises 315,648 observations, spanning a period of exactly nine years: from January 1, 2008, at 00:15 to January 1, 2017, at 00:00. The system load is recorded at a high-frequency resolution of 15 minutes, with all timestamps localized to Central European Time (UTC+1).

2.1. Temporal Alignment and Timezone Handling

A criticality in processing the raw time series was the presence of structural anomalies induced by the biannual Daylight Saving Time (DST) transitions. Because the data was recorded in local time, the raw dataset inherently contained missing observations during the spring transition (when clocks shift forward) and duplicated timestamps during the autumn transition (when clocks shift backward). To systematically resolve these inconsistencies, we leveraged the vectorized datetime handling capabilities of the "pandas" library in Python. By explicitly localizing the temporal index to the "Europe/Warsaw" timezone, we applied programmatic rules to resolve DST anomalies: ambigu-

ous overlapping hours during the autumn transition were algorithmically inferred based on chronological sequence, while nonexistent timestamps resulting from the spring shift were systematically realigned forward. Establishing this strictly chronological, timezone-aware temporal grid was a crucial foundational step to ensure the physical validity of the subsequent detrending operations and the accurate extraction of seasonal profiles.

2.2. Long-term Trend Extraction via Empirical Mode Decomposition (EMD)

To isolate the deterministic long-term variations from the high-frequency electricity demand, we employed Empirical Mode Decomposition (EMD). Unlike traditional frequency-based filters, EMD is an adaptive, data-driven technique capable of decomposing non-linear and non-stationary signals into a finite set of Intrinsic Mode Functions (IMFs) and a final monotonic residue. In our context, this residue fundamentally represents the underlying global trend of the power system's load.

To comprehensively illustrate the mechanics of the EMD algorithm and its effectiveness in separating distinct frequency bands, the initial decomposition was performed on a localized, representative subset of the time series. This approach allows for a detailed visualization of the individual IMFs—which intuitively capture the high-frequency noise and cyclical daily routines—alongside the local residual trend, avoiding the visual clutter associated with plotting the entire dataset (see Figure 1).

However, applying the standard EMD algorithm point-by-point to the entire high-frequency dataset is computationally prohibitive. To efficiently extract the historical long-term trend spanning the entire 2008–2016 period, an optimized data aggregation strategy was implemented. The raw 15-minute time series was first resampled to a daily frequency by computing the mean electrical load for each 24-hour period. This dramatically reduced the computational load while successfully preserving the macroeconomic dynamics of the power demand.

The EMD was then applied to this daily aggregated series, successfully isolating the final residue as the underlying macroeconomic trend. To seamlessly subtract this low-frequency trend from the original high-frequency demand, the daily trend was projected back onto the native 15-minute timeline. This upsampling was achieved via time-

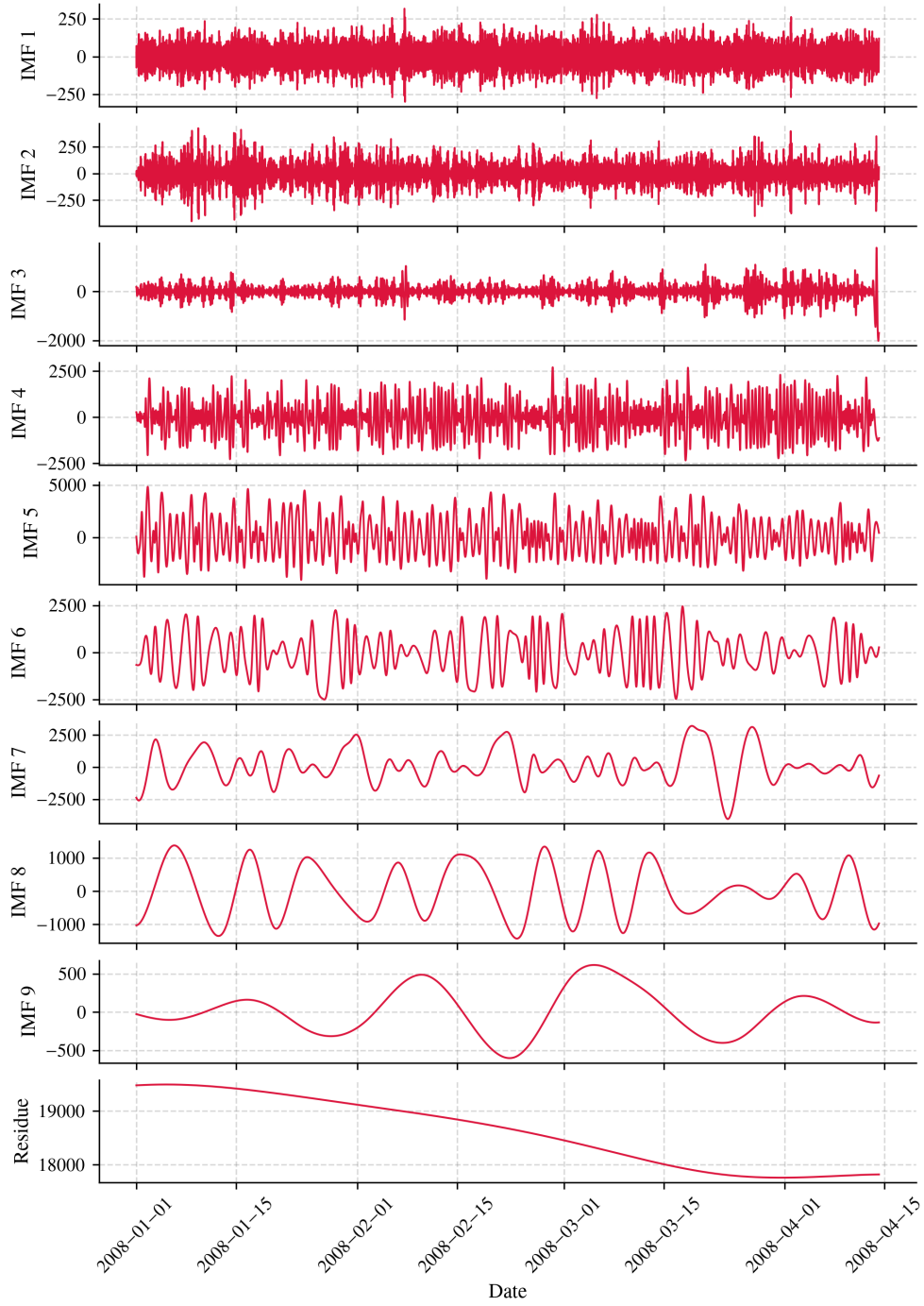
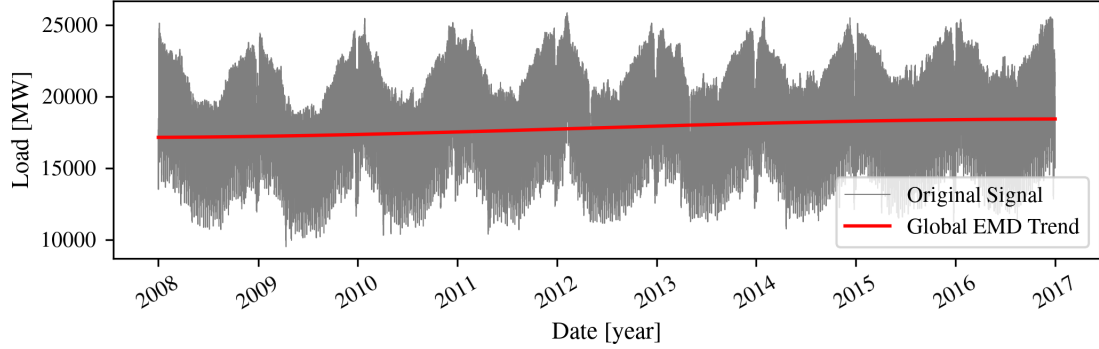
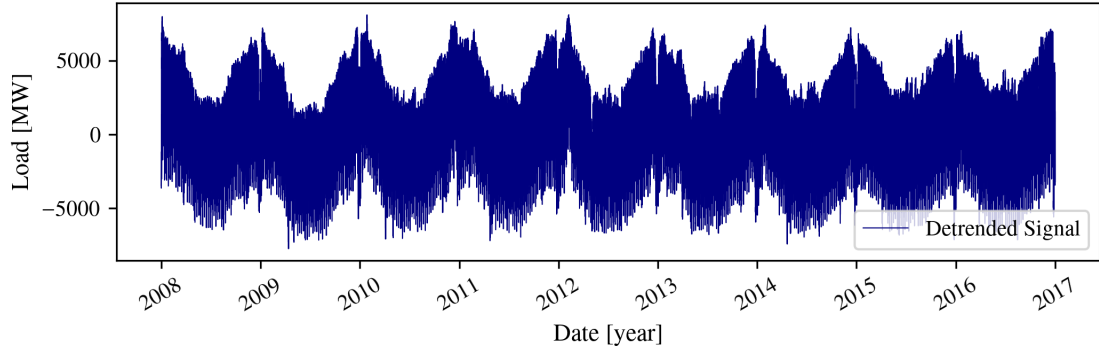


Figure 1: Empirical Mode Decomposition (EMD) applied to a representative subset of the Polish electricity demand (January to April 2008). The adaptive algorithm decomposes the raw time series into nine Intrinsic Mode Functions (IMFs), ordered from the highest-frequency stochastic variations to the slowest macro-dynamics, concluding with a monotonic "residue" that isolates the long-term structural trend of the power grid.



(a) Raw historical load of the Polish power system.



(b) Zero-mean detrended signal after EMD extraction.

Figure 2: Comparison of the electricity demand before and after the long-term trend extraction.

weighted linear interpolation, utilizing constant forward and backward filling to manage boundary conditions at the extreme edges of the dataset.

Finally, this continuous high-resolution trend line was subtracted from the raw system load, yielding a stationary, zero-mean stochastic baseline, which ensures optimal mathematical conditions for the subsequent Spectral Analysis (FFT) and SARIMA modeling. A visual comparison between the raw historical load (see Figure 2a) and the final zero-mean, detrended time series (see Figure 2b) clearly demonstrates the successful isolation of the long-term structural drift, leaving a stationary baseline characterized purely by seasonal and stochastic fluctuations.

3. Spectral Analysis and Deseasonalization

Once the long-term trend is removed, the remaining signal is dominated by powerful periodic oscillations rooted in human socio-economic behavior. To transition from a raw time series to a predictable process, we must first identify and isolate these deterministic seasonal components. This section details the two-step methodology employed: first, using spectral analysis to objectively identify the dominant frequencies, and second, applying a "folding" technique to extract a typical consumption profile.

3.1. Frequency Domain Analysis via Fast Fourier Transform (FFT)

Recent literature demonstrates that decoupling periodicities via spectral methods significantly enhances the performance of linear autoregressive

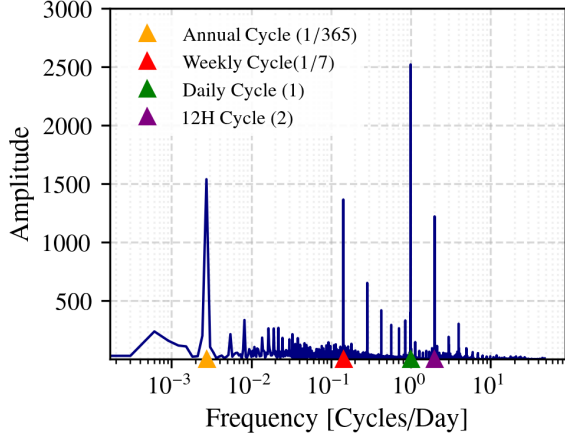


Figure 3: Fast Fourier Transform (FFT) amplitude spectrum of the detrended load signal, highlighting the dominant seasonal periodicities.

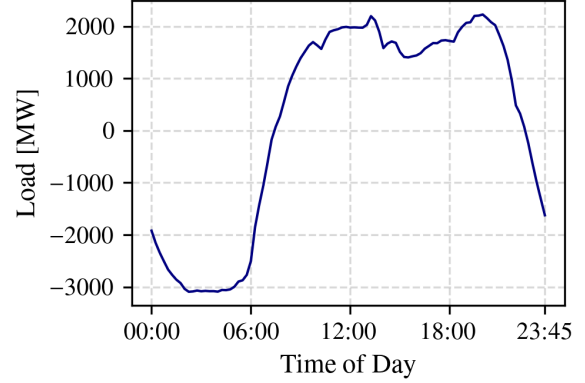
models [Karamolegkos and Koulouriotis, 2025]. To rigorously quantify the cyclical patterns within the detrended data, we transitioned the analysis from the time domain to the frequency domain using the Fast Fourier Transform (FFT). The resulting power spectrum allows us to identify the "harmonics" of the Polish power system—the specific frequencies where the signal's energy is most concentrated.

As illustrated in Figure 3, the periodogram reveals massive, distinct peaks that align perfectly with anthropogenic routines. The primary peak occurs at the 24-hour mark (daily cycle), followed by a significant peak at 168 hours (the weekly cycle). Secondary harmonics are also visible (e.g., at 12 hours), representing the characteristic "double-peak" structure of daily demand.

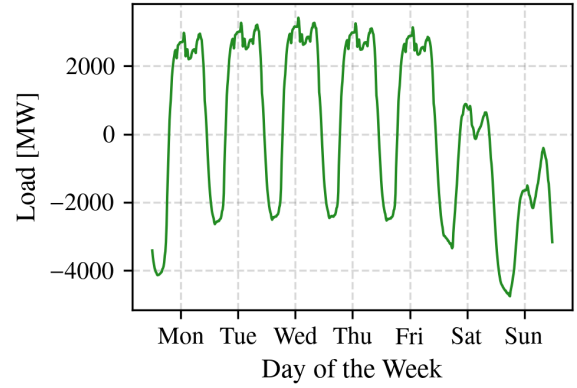
3.2. Extraction of the Typical Profile (Folding)

The spectral analysis confirmed that the Polish electricity demand is governed by rigid, hierarchical periodicities. To isolate these deterministic cycles, we applied a folding technique, which involves collapsing the multi-year dataset into a single representative temporal template by averaging corresponding intervals. We analyzed this phenomenon across three distinct scales to validate the consistency of the human-driven signal:

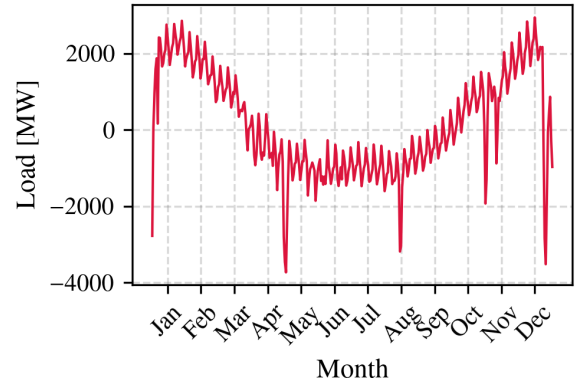
- **Daily Scale.** The extracted profile exhibits a classic bimodal distribution (see Figure 4a). We observe a first peak during the morning



(a) Daily Profile.



(b) Weekly Profile.



(c) Annual Profile.

Figure 4: Deterministic seasonal templates extracted via data folding. From top to bottom: (a) Daily bimodal routine, (b) weekly profile capturing the weekend load drop, and (c) Annual macro-seasonality highlighting summer valleys.

hours (09:00–13:00) associated with the ramp-up of industrial and commercial activities, followed by a second, more pronounced peak in the evening (19:00–22:00) driven by residential consumption and lighting. This "double-peak" structure is the time-domain manifestation of the 12-hour secondary harmonic previously identified via FFT.

- **Weekly Scale.** The weekly profile highlights the profound impact of socio-economic routines on power load (see Figure 4b). Consumption levels remain high and stable from Monday to Friday, while a significant structural drop is recorded during the weekend (Saturday and Sunday), reflecting the cessation of major industrial operations and the shift in public activities.
- **Annual Scale.** On a broader scale, the monthly averages reveal a marked decrease in electricity demand during spring and summer, particularly between April and August. (see Figure 4c). In the context of the Polish grid for the studied period, this suggests that the demand for summer cooling was not sufficient to offset the reduction in industrial output and the decreased need for artificial lighting during longer daylight hours. Furthermore, the annual profile exhibits distinct, sharp valleys throughout the year. These localized drops in power load directly correspond to major national and religious holidays—such as the extended May weekend (Labor Day on May 1 and Constitution Day on May 3), the movable feast of Corpus Christi in late spring, the November holidays (All Saints’ Day on Nov. 1 and Independence Day on Nov. 11), and the Christmas period (Dec. 25–26). During these exact windows, large-scale industrial operations and commercial activities are temporarily suspended.

Among these components, the weekly profile is the most critical for our modeling strategy. However, rather than relying on a single universal template (as the one shown in Figure 4b), this profile was computed conditionally based on the meteorological season to account for macro-level climatic and economic shifts. Since it captures the full variance of both intra-day fluctuations and the weekday-weekend transition, we utilized this season-specific template as the master deterministic component for

the deseasonalization process. By subtracting this "clockwork" from the detrended signal, we successfully suppressed the high-amplitude seasonal waves (see Figure 5). This operation allowed us to isolate the stationary stochastic residuals Δ , which capture the remaining short-term dynamics and temporal correlations, serving as the necessary input for the subsequent SARIMA forecasting framework.

4. Hybrid Forecasting Strategy

Having successfully isolated the strictly stationary, zero-mean residuals from the deterministic trend and periodic profiles, the final phase of our hybrid framework involves modeling these short-term autocorrelated fluctuations. To achieve this, we employed a Seasonal Autoregressive Integrated Moving Average (SARIMA) model, which is highly effective in capturing both localized serial correlations and residual seasonal dependencies.

4.1. Parameter Identification and Model Selection

The parameter selection process was conducted in two distinct phases: an initial visual diagnostic on the original high-frequency data, followed by an optimized algorithmic search on a resampled dataset. First, initial diagnostic evaluations were conducted by visually inspecting the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the residuals at their native 15-minute resolution (see Figure 6). These two correlograms serve complementary geometric purposes in time series analysis:

- **The ACF:** This function evaluates the overall correlation between the current signal and its past observations (lags), capturing both direct and cascading influences. By revealing how the echo of a past event continuously propagates through time, the ACF provides the necessary insight to determine the Moving Average (q) component of the model.
- **The PACF:** In contrast, the PACF isolates the pure, direct relationship between the signal and a specific past lag by mathematically filtering out the overlapping effects of all intermediate steps. This targeted metric is essential for identifying the Autoregressive (p) component, as it clearly indicates exactly how many prior observations carry unique predictive power for the next step.

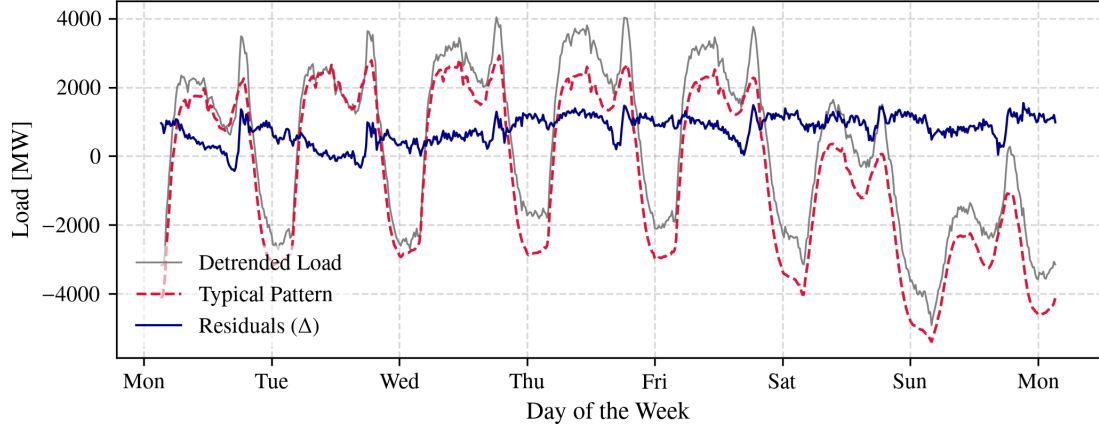


Figure 5: Deseasonalization process for a sample week (April 14–21, 2008). The plot demonstrates the isolation of stochastic residuals (blue line), obtained by subtracting the deterministic weekly profile (red dashed line) computed for the current season from the EMD-detrended load (grey line).

Upon visual inspection of the short-term lags, the PACF plot (Figure 6b) exhibits a sharp, significant cut-off after the second lag, strongly suggesting an optimal autoregressive order of $p = 2$. Concurrently, the ACF plot (Figure 6a) displays significant correlations at early lags before gradually decaying, pointing towards a moving average component of $q = 1$.

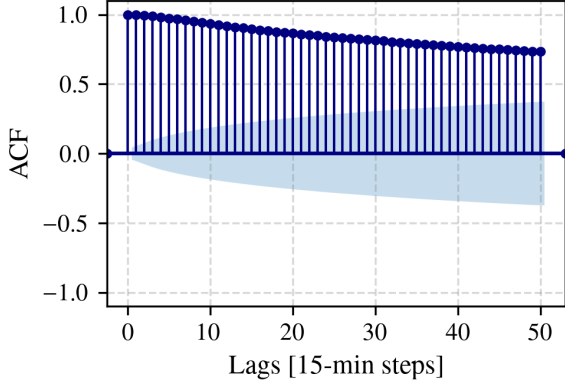
While this visual inspection provided a robust preliminary hypothesis for the non-seasonal architecture, training a maximum-likelihood SARIMA model on high-frequency time series over an extended training window is computationally prohibitive. Therefore, prior to initiating a rigorous algorithmic parameter search, the isolated residuals were resampled to a 30-minute resolution. Based on our prior spectral analysis (FFT), the fundamental daily seasonal cycle for this resampled modeling phase was mathematically defined by a period of $s = 48$ half-hour intervals. Although the deterministic macro-seasonality was previously removed via folding, this 24-hour cycle was explicitly maintained in the SARIMA structure to capture the residual serial correlations of daily stochastic shocks (e.g., an anomalous consumption deviation at 10:00 AM yesterday inherently influences the forecast error at 10:00 AM today).

To ensure an objective and reproducible parameter selection process, we implemented an algorithmic grid search utilizing the Stepwise Auto-ARIMA methodology. The hyperparameter search space

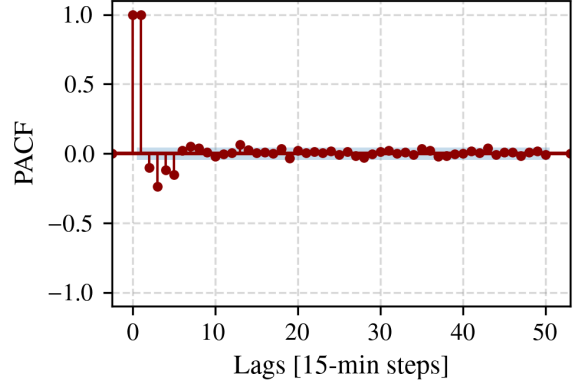
was deliberately bounded based on our prior analytical steps: both the non-seasonal and seasonal differencing orders (d and D) were strictly constrained to 0, as the input residuals had already been rendered perfectly stationary. The stepwise algorithm iteratively evaluated multiple parameter combinations, bounding the maximum non-seasonal orders to $p, q \leq 2$ and the seasonal orders to $P, Q \leq 1$ to prevent model overfitting. The optimization objective was the minimization of the Akaike Information Criterion (AIC). The algorithmic search concluded successfully, identifying the absolute optimal configuration as $\text{SARIMA}(2, 0, 1)(1, 0, 1)_{48}$. This mathematical result perfectly corroborated our initial visual estimations, explicitly confirming that the stochastic innovations in the Polish power grid are best predicted by a combination of the two immediately preceding half-hour observations ($p = 2$), one past forecast error ($q = 1$), and a single seasonal lag corresponding to the exact same time of the previous day ($P = 1, Q = 1$ at lag 48).

4.2. Signal Reconstruction

The fundamental strength of the proposed methodology lies in its hybrid nature: the deterministic elements of the power grid (macro-economic evolution and human routines) are modeled independently from its stochastic behavior. Once the SARIMA model generates the forecasted residuals for the target horizon, the true predictive power of the system is realized by recombining the disparate



(a) Auto-Correlation Function (ACF)



(b) Partial Auto-Correlation Function (PACF)

Figure 6: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) computed on a representative subset comprising the last 2,000 observations of the stationary residuals.

components into the original physical domain. The final forecasted load ($\widehat{Load}_t$) is obtained through the following additive reconstruction formula:

$$\widehat{Load}_t = Trend_t + Profile_t + \widehat{\Delta}_t \quad (1)$$

where the subscript t refers to the specific time step of the prediction and the "hat" notation ($\widehat{\cdot}$) is strictly reserved for estimated quantities carrying predictive uncertainty, which in this framework originate solely from the forecasted SARIMA residuals. The individual components of the hybrid forecasting equation are defined as follows:

- $Trend_t$ (The Macro-economic Baseline): This component, extracted via EMD, represents the long-term evolution of power consumption. Given its inherently low-frequency nature, it is treated as a deterministic element during the out-of-sample forecasting phase. For the 15-day horizon evaluated in this study, it is extrapolated using a naive approach, freezing its future projection as a constant equal to the last available observation of the training set.
- $Profile_t$ (The Deterministic Seasonality): This is the fixed season-specific weekly template obtained via the folding technique. To rigorously prevent any forward-looking bias (data leakage), this temporal averaging is computed exclusively over all historical data available up to the forecasting cutoff point (i.e., the

end of the training phase). Crucially, this component is intentionally modeled as a purely deterministic framework, effectively ignoring the variance of its vast historical source distribution to provide a rigid, cyclical baseline of human routines for the out-of-sample horizons.

- $\widehat{\Delta}_t$ (The Modeled Stochastic Fluctuation): This is the output of the SARIMA model. It captures the short-term deviations and auto-correlated dynamics of the grid that remain after filtering out the deterministic periodic profile and the long-term trend.

4.3. Results and Evaluation

To empirically evaluate the predictive capabilities of the hybrid forecasting architecture, the model was tested on a 15-day out-of-sample forecasting horizon. This specific temporal window was selected to ensure a stable operational period, strictly devoid of macro-seasonal transitions and disruptive national holidays. As established during the SARIMA parameter identification phase (Section 4.1), both the signal reconstruction and the final performance evaluation were conducted at the stable 30-minute aggregated resolution. The SARIMA model was trained on the historical residuals up to the end of the training window, and a multi-step-ahead forecast was generated. To quantify the absolute accuracy of the point forecasts, we utilized the Root Mean Squared Error (RMSE), calculated by comparing the final reconstructed pre-

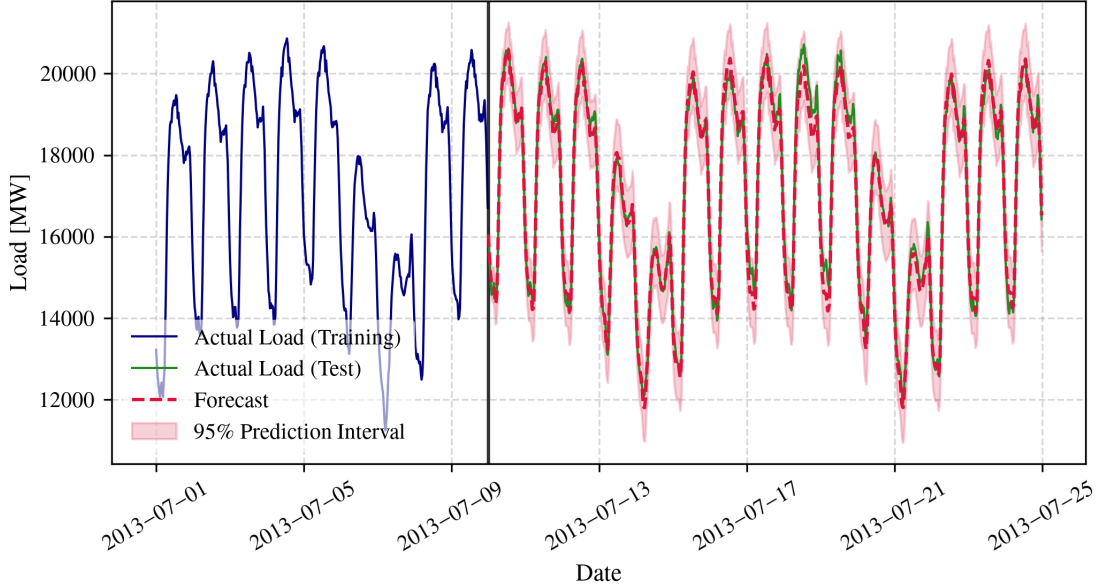


Figure 7: The plot displays the reconstructed forecasted signal generated by the hybrid method (red dashed line). The SARIMA model was trained on historical residuals from June 10 to July 9, 2013 (blue solid line), and evaluated on an out-of-sample test window from July 10 to July 24, 2013 (green solid line). The red shaded area represents the 95% prediction interval.

dictions against the actual physical load observed in the resampled test set. Furthermore, alongside the point forecast, we extracted the 95% Prediction Intervals ($\alpha = 0.05$) generated by the maximum-likelihood estimation of the SARIMA algorithm. To translate these bounds back into the real-world scale, the deterministic components ($Trend_t$ and $Profile_t$) were added to both the lower and upper statistical limits of the residuals.

A critical observation regarding the reconstructed prediction bounds is their behavior over time. In traditional forecasting models applied directly to raw, non-stationary time series (such as standard ARIMA models with $d > 0$), the forecast variance inherently grows indefinitely, causing the intervals to diverge and become practically useless as the horizon expands. Conversely, in our hybrid framework, the forecast error does not diverge to infinity. Because the SARIMA algorithm operates exclusively on strictly stationary, zero-mean residuals, the forecast inherently mean-reverts. As the forecasting steps move further into the future—beyond the "memory" of the autoregressive components—the predicted residuals smoothly converge to zero, and the width of the prediction intervals stabilizes, converging asymptotically to the

unconditional variance of the residual time series. This stabilization is not a limitation, but rather a strong validation of the pre-processing phase: it demonstrates that the non-stationary trends and explosive variance have been successfully isolated within the deterministic components (EMD and seasonal folding), leaving only a bounded, stationary signal to be modeled probabilistically.

4.4. Model Limitations and Performance in Anomalous Conditions

While the hybrid EMD-SARIMA framework demonstrates high accuracy during "standard" operational weeks, its predictive performance is inherently constrained by the assumption of socio-economic and environmental stationarity. The model relies on a deterministic season-specific weekly profile that represents an "average" behavior, and a low-frequency trend that is assumed static over short forecasting horizons. Consequently, its accuracy diminishes significantly when the power system is subjected to exogenous shocks, non-routine events, or extended temporal windows that span across seasonal transitions.

To illustrate these boundaries, we analyze specific case studies where the model's fundamental

assumptions fail to align with the observed physical reality.

4.4.1. Limitations in Long-Term Forecasting Horizons

The proposed architecture is specifically designed and optimized for Short-Term Load Forecasting (STLF). To rigorously evaluate the temporal boundaries of the model, an extended 60-day out-of-sample horizon (September-October 2009) was tested. In this scenario, the macroeconomic trend extracted during the training phase (August 2009) is frozen and extrapolated as a constant.

As depicted in Figure 8, while the inclusion of seasonal profiles prevents the actual load from breaching the 95% prediction intervals, a closer inspection reveals a lingering limitation during structural seasonal transitions. Although the forecast remains strictly bounded and mathematically stable, the predicted mean exhibits a largely stationary behavior over the extended horizon. In contrast, the actual load (green line) demonstrates clear low-frequency, long-term variations, reflecting the gradual shift in energy demand as autumn progresses.

This visual evidence highlights a crucial modeling insight: during prolonged seasonal transitions, merely incorporating the current season's periodic profile is insufficient. To perfectly track these prolonged shifts, the deterministic macro-trend must also be treated as a dynamic, non-stationary component rather than a static baseline. Therefore, while the stationary SARIMA residuals successfully prevent the mathematical divergence of the confidence bounds, extending this framework to highly accurate Long-Term Load Forecasting (LTLF) applications would strictly require dynamic trend extrapolation techniques or the integration of exogenous macroeconomic or weather-related variables.

4.4.2. The Impact of National Holidays: Christmas 2008

As shown in Figure 9, the model exhibits a severe overestimation of the electrical load during the last week of December 2008. In this period, the deterministic weekly profile—which expects a standard working-day routine—fails to account for the massive suspension of industrial and commercial activities typical of the Christmas holidays. Since the SARIMA component only manages short-term stochastic innovations, it cannot compensate for

such a structural, calendar-driven shift in the baseline. This result highlights the necessity of integrating "holiday variables" or a dedicated holiday-aware profile to handle non-typical days.

4.4.3. Extreme Meteorological Events: The 2012 Cold Wave

A different type of failure is observed during periods of extreme weather, such as the severe European cold wave of February 2012 (see Figure 10). In this instance, the real load exceeds the forecast; nevertheless, it still lies in the 95% confidence interval. This underestimation occurs because the model lacks exogenous inputs for temperature. During exceptionally cold winters, the demand for heating increases non-linearly, shifting the entire consumption curve upward. While the SARIMA component attempts to adjust for this "shock" through its autoregressive terms, the magnitude of the thermal-driven demand far exceeds the statistical memory of the residuals, leading to a persistent forecasting bias.

5. Conclusions

In this study, we presented a hybrid framework for the analysis and forecasting of electricity demand in the Polish power system. By decoupling the signal into its deterministic and stochastic components, the proposed methodology addresses the non-stationarity and multi-scale seasonality characteristic of high-frequency energy data. Our methodological approach yielded the following key outcomes:

- **Trend Decoupling:** As detailed in Section 2.2, Empirical Mode Decomposition (EMD) was applied to isolate the non-linear, long-term macroeconomic trend without relying on rigid parametric assumptions.
- **Deseasonalization via Folding:** By employing the folding technique described in Section 3.2, a deterministic season-specific weekly profile was extracted to represent the socioeconomic routines of the Polish population. This allowed for the separation of the "clockwork" of human activity from the underlying stochastic noise.
- **Hybrid Synthesis and Model Boundaries:** As formalized in Section 4, the

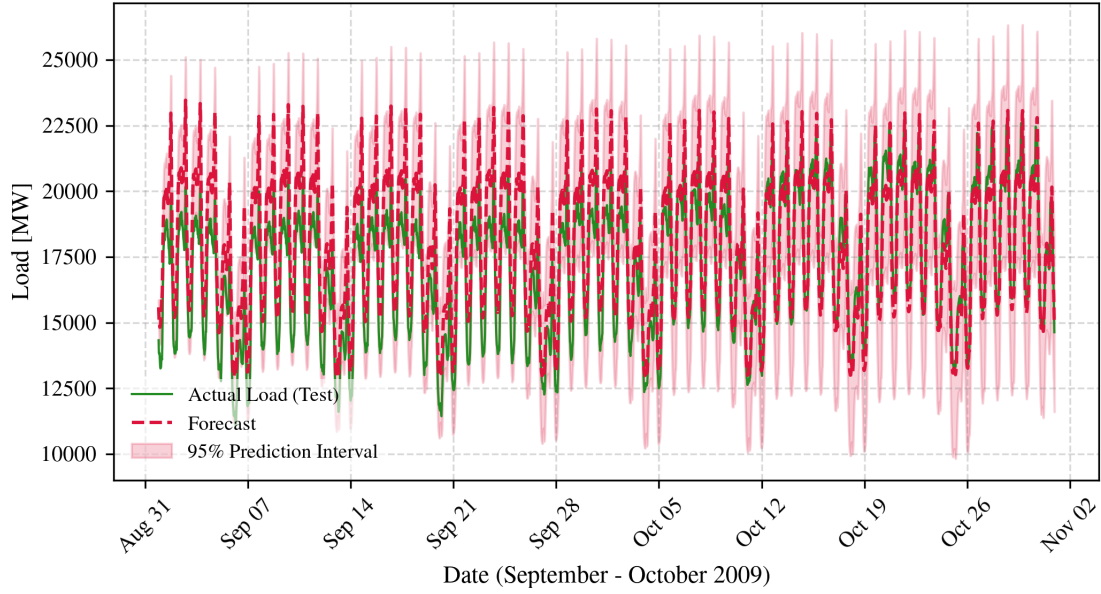


Figure 8: Long-term forecasting evaluation over a 60-day out-of-sample horizon (September–October 2009). The model is trained exclusively on August data, representing a historical seasonal minimum. While the stationary SARIMA residuals successfully maintain stable 95% prediction intervals—keeping the actual load securely within the bounds—the point forecast remains rigidly locked to the static summer baseline. As industrial activity resumes during autumn, the actual load exhibits non-stationary upward variations. This visual evidence confirms that while the stochastic component is properly bounded, accurate Long-Term Load Forecasting (LTLF) applications require dynamic trend extrapolation.

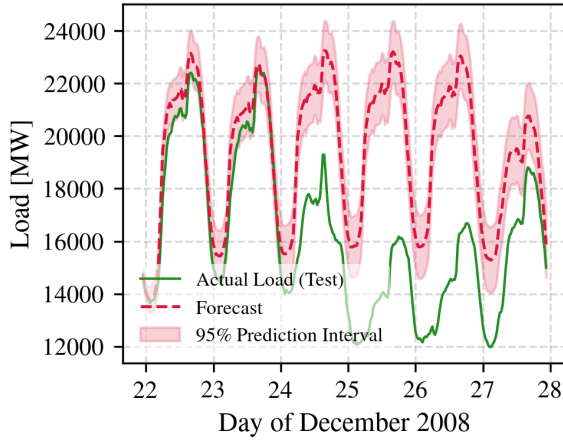


Figure 9: Out-of-sample forecasting performance during the Christmas holidays of December 2008. The plot highlights a structural overestimation of the electrical load, as the real demand (green line) consistently falls below the hybrid prediction (red dashed line) and breaks through the lower 95% confidence bounds.

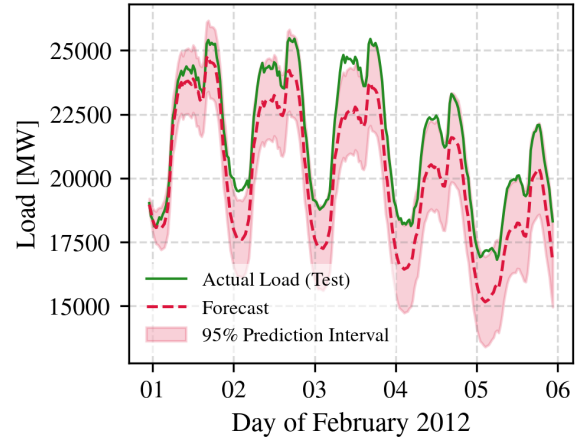


Figure 10: Out-of-sample forecasting performance during the severe European cold wave of February 2012. The plot highlights a structural underestimation of the electrical load, as the real demand (green line) exceeds the hybrid prediction (red dashed line), eventually breaching the 95% confidence bounds.

SARIMA model was trained exclusively on the resulting stationary residuals. The evaluation indicated that predicting the underlying component independently from large-scale trends yields stable forecasting bounds, reflecting the successful stationarization of the input signal. However, the analysis presented in Section 4.4 highlighted critical methodological boundaries. First, the reliance on a static deterministic profile increases the model’s sensitivity to calendar-driven shifts and exogenous environmental shocks, as observed during the Christmas holidays and the 2012 extreme cold wave. Second, evaluating the framework over an extended 60-day out-of-sample horizon revealed that although the model remains mathematically stable—successfully containing the actual load within its prediction intervals—its point-forecast accuracy noticeably diminishes. Freezing the low-frequency EMD trend at the training season’s baseline introduces a structurally stationary behavior that fails to adapt to real grid variations during macro-seasonal transitions. Ultimately, this confirms that while the proposed architecture is highly effective and robust for Short-Term Load Forecasting (STLF), its direct application to Long-Term Load Forecasting (LTLF) strictly requires dynamic trend extrapolation or the integration of macroeconomic and weather-related drivers.

Consequently, we conclude that while the hybrid EMD-SARIMA architecture provides a robust baseline for national load forecasting under standard conditions, its current reliance on static trend extrapolation limits its reliability over extended horizons. Future iterations should integrate weather-related covariates and holiday-aware logic, such as semi-parametric additive structures [Fan and Hyndman, 2012], to better manage non-deterministic volatility. Furthermore, to effectively transition toward Long-Term Load Forecasting (LTLF), subsequent research should explore dynamic trend modeling, either through trend regression techniques or by implementing a secondary SARIMA model specifically optimized to capture and extrapolate the low-frequency macroeconomic baseline.

Code Availability

The source code for the hybrid EMD-SARIMA architecture, along with the Jupyter Notebook used for the analysis and visualization, is openly available in the following GitHub repository: [github repository](#).

References

- S. Fan and R. J. Hyndman. Short-term load forecasting based on a semi-parametric additive model. *IEEE Transactions on Power Systems*, 27(1):134–141, 2012. URL <https://robjhyndman.com/papers/2010STLF-FinalR1.pdf>.
- S. Karamolegkos and D. Koulouriotis. Advancing short-term load forecasting with decomposed fourier arima: A case study on the greek energy market. *Energy*, 325:130605, 2025. URL <https://ideas.repec.org/a/eee/energy/v325y2025ics0360544225014963.html>.
- K. Misiurek, T. Olkuski, and J. Zyśk. Review of methods and models for forecasting electricity consumption. *Energies*, 18(15):4032, 2025. doi: 10.3390/en18154032. URL <https://www.mdpi.com/1996-1073/18/15/4032>.
- Polskie Sieci Elektroenergetyczne (PSE). Historical electricity load data of the polish power system (2008–2016). <https://www.pse.pl/web/pse-eng/data/polish-power-system-operation/load-of-polish-power-system>, 2017. National Transmission System Operator. Open Data Repository. Accessed: 2026.
- M. Zielińska-Sitkiewicz, M. Chrzanowska, and K. Furmańczyk. Analysis of electricity consumption in poland using prediction models and neural networks. *Energies*, 14(20):6619, 2021. doi: 10.3390/en14206619. URL <https://www.mdpi.com/1996-1073/14/20/6619>.